# Text 1

第 1 段

The grammar school boy from Stratford-upon-Avon has landed a scholarly punch after ground-breaking research showed that Shakespeare does benefit children's literacy and emotional development. But only if you act him out.

直译：这位出身埃文河畔斯特拉特福文法学校的男孩，如今在学术争论中打出了一记有力的拳头：突破性研究表明，莎士比亚的作品确实有益于儿童的读写能力与情感发展。但前提是，你得把他的作品演出来。

第 2 段

A study found that a "rehearsal room" approach to teaching Shakespeare broadened children's vocabulary and the complexity of their writing as well as their emotional literacy. "The research shows that the way actors work makes a big difference to the way children use language and also how they think about themselves," Jacqui O'Hanlon of the Royal Shakespeare Company (RSC), which commissioned the study, said.

直译：一项研究发现，以“排练室”方式教授莎士比亚，扩大了儿童的词汇量，提高了写作的复杂程度，也提升了他们的情感理解与表达能力。委托开展这项研究的皇家莎士比亚剧团（RSC）的杰奎·奥汉隆说：“研究表明，演员工作的方式，会显著影响儿童如何运用语言，以及如何看待自己。”

第 3 段

The randomised control trial involved hundreds of year 5 pupils—aged nine and ten—at 45 state primary schools that had not been "previously exposed to RSC pedagogy". They were split into target and control groups and asked to write, for example, a message in a bottle as Ferdinand after the shipwreck in The Tempest. The target group was given a 30-minute drama-based activity to accompany the passage.

直译：这项随机对照试验涉及四十五所公立小学数百名九至十岁的五年级学生；他们此前都“未接触过 RSC 教学法”。学生被分成实验组和对照组，并被要求完成写作任务，例如扮演《暴风雨》中遭遇海难后的费迪南德，写一封漂流瓶里的信。实验组还参加了一个与该段文本配套、时长三十分钟的戏剧活动。

第 4 段

The peer-reviewed results showed that the target group of pupils drew on a wider vocabulary, used words "classed as more sophisticated or rarer", and wrote at greater length. They also "appear to be more comfortable writing in role...while [control] group imagine how they themselves would react to being shipwrecked, [target] group put themselves in the shoes of a literary character and express that character's emotion".

直译：经过同行评审的结果显示，实验组学生调用了更广的词汇，使用了“被归类为更复杂或更少见”的词语，而且写得更长。他们似乎也“更自如地以角色身份写作……对照组想象的是自己遭遇海难会如何反应，而实验组则设身处地站在一个文学人物的位置，表达那个角色的情绪”。

第 5 段

The Time to Act study also found that while control pupils relied on “desert island clichés” such as palm trees, target pupils were “more expansive [giving] a broader picture of the sky, the sea and the atmospheric conditions”.

直译：《行动的时刻》研究还发现，对照组学生依赖棕榈树之类的“荒岛陈词滥调”，而实验组学生的描写“更为丰富，呈现了天空、大海和天气状况更广阔的画面”。

第 6 段

O’Hanlon said she had been most surprised by the “emotional literacy that was evident in the [target] children’s writing” and that they were “more resilient in their writing, more hopeful”. She added: “The emotional understanding was very evident and it is probably related to the [rehearsal room process] where you are used to trying to imagine your way through. They were comfortable in describing different emotional states and part of what you do in drama is put yourself in different shoes.” The study showed the importance of embedding the arts in education, she said.

直译：奥汉隆说，最令她惊讶的是，“实验组儿童写作中明显表现出的情感素养”，以及他们“在写作中表现得更坚韧、更有希望”。她补充说：“情感理解十分明显，这可能与排练室过程有关；在那个过程中，你习惯于凭借想象进入并经历情境。他们能够自如地描述不同的情绪状态，而戏剧活动的一部分，就是设身处地扮演不同的人。”她说，该研究显示了把艺术融入教育的重要性。

第 7 段

But could the results be replicated with any old dramatist? O’Hanlon said more research would be needed but suggested that Shakespeare’s use of 20,000 words, compared with the everyday 2,000 words, gave a “massive expansion of language into children’s lives”, which was combined with children “using their whole bodies to bring words to life”.

直译：但是，换成任何其他剧作家，能否复制这样的结果？奥汉隆说，还需要更多研究，但她认为，与日常使用的两千个词相比，莎士比亚使用的两万个词，把“极其丰富的语言拓展带入儿童生活”；而这又与儿童“运用整个身体让词语活起来”的过程结合在一起。

# Text 2

第 1 段

I was shocked to learn recently that some scientists want to scale back their research in an effort to decrease carbon emissions. The crisis is here, they said, and we need to cut back on our energy-intensive modelling. At the very least, we need to make our energy use far more sustainable.

直译：最近得知一些科学家希望缩减研究，以减少碳排放，我感到震惊。他们说，危机已经到来，我们需要减少耗能密集的建模工作。至少，我们需要让能源使用方式变得可持续得多。

第 2 段

It is unarguable that our laboratories, scientific instruments, rockets and satellites—the tools we scientists need to measure the planet's pulse—demand significant amounts of energy both in their construction and operation. And it is equally true that science's unrelenting appetite for information has caused a mushrooming of energy-intensive data centres around the world. According to the International Energy Agency, these buildings now consume about 1 per cent of the world's electricity.

直译：无可争辩的是，我们的实验室、科学仪器、火箭和卫星——这些科学家用来测量地球脉搏的工具——无论在建造还是运行时，都需要大量能源。同样真实的是，科学对信息永不停歇的渴求，促使耗能密集的数据中心在全球迅速增加。根据国际能源署的数据，这些建筑如今消耗全球约百分之一的电力。

第 3 段

However, this is a price we must pay for understanding the world. How can we inform decision makers about the best ways to bring down carbon emissions if we can't track the amount of carbon dioxide in the atmosphere, where it's coming from and who's producing it? The carbon emissions from technological research are well spent: ultimately this research will safeguard the future of our planet.

直译：然而，这是我们为理解世界必须付出的代价。如果我们无法追踪大气中二氧化碳的含量、它来自哪里、由谁产生，又怎能告知决策者降低碳排放的最佳方式？技术研究所产生的碳排放，是值得付出的代价：最终，这些研究将守护地球的未来。

第 4 段

It can be hard for scientists to make the case because our work is complex, often takes place behind closed doors and does not always lend itself to easy interpretation or explanation. But demonstrating the efficacy of science will be crucial if we are to solve humanity's greatest challenges.

直译：科学家可能很难把这一道理说清楚，因为我们的工作复杂，常在公众看不到的地方进行，而且并不总是容易解释或说明。但是，如果要解决人类最重大的挑战，展示科学的成效将至关重要。

第 5 段

Recognising the hope that science and engineering can bring was the impetus behind the creation of the Millennium Technology Prize, which is now entering its 20th year as a celebration of human ingenuity. One of the past winners, Professor Martin Green from the University of New South Wales, Australia, is the inventor of the Passivated Emitter and Rear Cell technology which is now found in most of the world's solar panels. Thanks to his invention, we have a real chance to decrease the world's carbon emissions.

直译：认识到科学与工程能够带来的希望，正是设立千禧年技术奖的推动力。如今，这个赞扬人类创造才智的奖项正进入第二十个年头。往届获奖者之一、澳大利亚新南威尔士大学的马丁·格林教授，是钝化发射极和背面电池技术的发明者；如今，世界上大多数太阳能电池板都采用了这一技术。多亏他的发明，我们才真正有机会降低全球碳排放。

第 6 段

Every day, scientists, technologists and engineers are discovering new ways to exploit renewable energy sources and develop techniques not just to use power more intelligently but to power our intelligence. A great example of this is Europe's largest supercomputer, LUMI in Finland, which is astonishingly carbon-negative. Established in an old paper mill, it is powered by a nearby river and its remote heat warms the people who live in the surrounding town of Kajaani.

直译：科学家、技术专家和工程师每天都在发现利用可再生能源的新方式，并开发技术，不仅让我们更聪明地使用能源，也为我们的智能活动提供能源。欧洲最大的超级计算机——位于芬兰的 LUMI——就是一个极佳的例子，它竟然实现了负碳排放。它设在一座旧造纸厂内，由附近河流供电，其热量被送往远处，为周边卡亚尼镇的居民供暖。

第 7 段

If the world is to meet its net-zero ambitions, we must think hard about how we can deliver sustainable computing and deliver more LUMIs.

直译：如果世界要实现净零排放的雄心，我们就必须认真思考，如何实现可持续计算，以及如何建设更多像 LUMI 这样的设施。

# Text 3

第 1 段

Ever since taking on Netflix Inc. at its own game, old Hollywood has struggled to turn a profit in streaming, with the likes of Disney+, Peacock and Paramount+ losing billions of dollars each year, sparking concerns on Wall Street that the services will never be as profitable as cable once was. But the age of streaming has been a boon for some unintended winners: pirates that use software to rip a film or television show in seconds from legitimate online video platforms and host the titles on their own, illegitimate services, which rake in about $2 billion annually from ads and subscriptions.

直译：自从开始在 Netflix 的主场与它竞争以来，传统好莱坞一直难以在流媒体业务中盈利。Disney+、Peacock 和 Paramount+ 等平台每年亏损数十亿美元，使华尔街担忧这些服务永远无法像过去的有线电视那样赚钱。但是，流媒体时代却使一些意料之外的赢家获益：盗版者利用软件，在几秒钟内从合法网络视频平台抓取电影或电视剧，再把这些作品放到自己的非法服务上。这些服务每年从广告与订阅中攫取约二十亿美元。

第 2 段

With no video production costs, illegal streaming sites have achieved profit margins approaching 90%, according to the Motion Picture Association (MPA), a trade group representing Hollywood studios that's working to crack down on the thousands of illegal platforms that have cropped up in recent years.

直译：代表好莱坞制片公司的行业组织美国电影协会（MPA）表示，由于没有视频制作成本，非法流媒体网站已实现接近百分之九十的利润率。该协会正努力打击近年来冒出的数千个非法平台。

第 3 段

Initially the rise of legitimate online businesses such as Netflix actually helped curb digital piracy, which had largely been based on file uploads. But now piracy involving illegal streaming services as well as file-sharing costs the US economy about $30 billion in lost revenue a year and some 250,000 jobs, estimates the US Chamber of Commerce's Global Innovation Policy Center. The global impact is about $71 billion annually.

直译：最初，Netflix 等合法网络企业的兴起，实际上帮助遏制了数字盗版，而当时盗版主要依靠文件上传。但美国商会全球创新政策中心估计，如今涉及非法流媒体服务及文件共享的盗版，每年使美国经济损失约三百亿美元收入，以及大约二十五万个工作岗位。其全球影响约为每年七百一十亿美元。

第 4 段

“The people who are stealing our movies and our television shows and operating piracy sites are not mom and pop operations,” says Charlie Rivkin, chief executive officer of the MPA. “This is organized crime.” Rivkin joined the MPA in 2017 after the organization failed five years earlier to build consensus between Hollywood and Silicon Valley to win passage of legislation in Congress aimed at stopping online piracy. In 2017 the association formed the Alliance for Creativity and Entertainment (ACE), an enforcement task force of about 100 detectives circling the globe to help local authorities arrest streaming pirates. ACE says it’s helped shrink the number of illegal streaming services in North America to 126, from more than 1,400 in 2018, aided in part by the MPA’s support for a 2020 federal law that made large-scale streaming of copyrighted material a serious crime.

直译：美国电影协会首席执行官查理·里夫金说：“那些窃取我们的电影和电视节目、经营盗版网站的人，并不是夫妻店式的小经营者。这是有组织犯罪。”里夫金于二〇一七年加入 MPA。五年前，该组织曾试图促成好莱坞与硅谷达成共识，以推动国会通过旨在阻止网络盗版的法律，但未能成功。二〇一七年，该协会成立了创意与娱乐联盟（ACE），这是一个由约一百名调查人员组成的执法特别小组，在全球活动，协助地方当局逮捕流媒体盗版者。ACE 表示，它已帮助把北美非法流媒体服务的数量，从二〇一八年的一千四百多个减少到一百二十六个。部分助力来自 MPA 对二〇二〇年一项联邦法律的支持；该法把大规模传播受版权保护内容的流媒体行为认定为严重犯罪。

第 5 段

Consulting firm Parks Associates predicts that legitimate US streaming services’ cumulative loss from piracy since 2022 will reach $113 billion in the next two years. “While there is some optimism that emerging countermeasures and best practices may see piracy begin to plateau by 2027, there is no consensus among stakeholders as to when it may begin to decline,” says analyst Steve Hawley.

直译：咨询公司 Parks Associates 预测，自二〇二二年起，美国合法流媒体服务因盗版造成的累计损失，将在未来两年达到一千一百三十亿美元。分析师史蒂夫·霍利说：“虽然有人乐观地认为，新出现的应对措施和最佳实践，可能让盗版到二〇二七年开始趋于平稳，但利益相关者对于它何时可能开始下降，尚无共识。”

# Text 4

第 1 段

Visit any antiques store and you'll encounter artifacts from the past: photographs, letters, a brochure detailing the Sinclair dinosaur exhibit from the 1964-1965 World's Fair, the ephemera of history. Yet these objects aren't truly ephemeral, because they're still here, decades, even centuries later. Why? Because they're tangible.

直译：走进任何一家古董店，你都会遇到过去的遗物：照片、信件、一份详细介绍一九六四至一九六五年世界博览会上辛克莱恐龙展的宣传册——这些历史中的零散短期印刷品。然而，这些物件并不真的转瞬即逝，因为几十年、甚至几百年之后，它们仍然存在。为什么？因为它们是有形的。

第 2 段

Have you pondered the life cycle of intangible formats, digital information, given that those who produce these artifacts seldom make provision for their long-term preservation? For millennia, we've known what we've known due to artifacts that have survived, often despite their original creators' neglect. The thing itself is the medium that delivers the information. At the time of creation, no attempts were made at intentional preservation, yet analog materials have a chance of surviving and serving as the historical record that biographers, historians, and novelists rely on. Libraries and archives have traditionally shouldered the responsibility of organization, preservation, and access to information. One of S.R. Ranganathan's foundational Laws of Library Science is "Save the time of the reader." Thus, librarians digitize the tangible so that researchers the world over can quickly search and access their holdings. The result is an embarrassment of historical riches, which brings its own needle-and-haystack problems.

直译：考虑到创造这些物件的人很少为它们的长期保存作出安排，你是否思考过无形形式，也就是数字信息的生命周期？几千年来，我们之所以知道自己所知道的事，是因为有一些物件保存了下来，而且往往是在原创造者疏于照料的情况下仍然幸存。物件本身就是传递信息的媒介。创作之时，人们并未有意尝试保存它们，但模拟形式的材料仍有机会存留下来，成为传记作者、历史学家和小说家赖以工作的历史记录。图书馆与档案机构传统上承担着信息组织、保存及提供利用的责任。S. R. 阮冈纳赞的《图书馆学五定律》中的一项基础原则是“节省读者的时间”。因此，图书馆员把有形资料数字化，使全世界的研究者能够迅速检索并获取馆藏。结果是，历史资料丰富得令人应接不暇，而这又带来了大海捞针式的检索难题。

第 3 段

Librarians' selfless devotion can act against us when users point to universality of access by holding up a cellphone and saying, "it's all in here" as evidence that libraries are less vital for researchers today. Yet how was that universality of access made possible and, perhaps more importantly, how is it maintained? Who curates what is preserved? When it comes to born-digital information, the terrifying answer can be: if not librarians and archivists, then no one. Digital information requires a great deal more care than analog.

直译：当用户举起手机，说“一切都在这里”，并把这种普遍可访问性当作图书馆如今对研究者已不那么重要的证据时，图书馆员的无私奉献反而可能对我们不利。然而，这种普遍可访问性究竟是怎样实现的？也许更重要的是，它又是怎样维持的？由谁选择并管理需要保存的内容？谈到原生数字信息时，令人恐惧的答案可能是：如果不是图书馆员和档案工作者，那就没有人。数字信息比模拟信息需要多得多的照料。

第 4 段

Even when a digital object is preserved, it may only be the carrier that's saved, not the information itself. As technology advances and a format becomes obsolete, the object is useless. Have you ever stared helplessly at a ZIP disk, thinking: how do I get the files off this? Without constant migration of digital assets, a nightmare about the foreseeable future is what keeps historians up at night: a historical record that abruptly stops when digital replaces analog.

直译：即使某个数字对象被保存下来，被保住的也可能只是载体，而不是信息本身。随着技术进步、某种格式过时，该物件就会变得无用。你是否曾无助地盯着一张 ZIP 磁盘，想着：我该怎样把里面的文件取出来？如果不持续迁移数字资产，关于可预见未来的一个噩梦就会让历史学家夜不能寐：当数字形式取代模拟形式时，历史记录却突然中断。

第 5 段

As a librarian whose day job revolves around special collections and digital assets, I share the night terrors of historians, and I'd be lying if I said a comprehensive preservation solution currently exists. Yet researchers can take some comfort in the fact that there are a multitude of librarians devoted to discovering, organizing, and preserving digital information for researchers current and future. Librarians are uniquely positioned to understand how end users seek and use information. Thus we play an integral role in identifying, preserving, and providing accessibility to digital artifacts so that, while future researchers may find the digital realm a challenging place to ply their trade, they won't find it an impossible one.

直译：作为一名日常工作围绕特藏与数字资产展开的图书馆员，我与历史学家有着同样的夜间恐惧。如果我说，目前已经存在一种全面的保存解决方案，那就是撒谎。不过，研究者可以从这样一个事实中获得些许安慰：有许多图书馆员，正致力于为现在和未来的研究者发现、组织并保存数字信息。图书馆员具有独特的条件，能够理解最终用户如何寻找和使用信息。因此，在识别、保存数字物件并提供访问方面，我们发挥着不可或缺的作用。这使未来研究者即使发现数字领域是一个颇具挑战性的工作场所，也不会发现它是一个根本无法工作的地方。